

## I. Decision Tree Limitations and Ensemble Motivation (1–10)

1. **Q:** What is the main disadvantage of a single, unregularized decision tree model?

**A:** It tends to **overfit** the training data and has **high variance**, leading to poor generalization.

2. **Q:** What structural issue makes a simple decision tree a "weak learner"?

**A:** Its predictions are **highly sensitive to the training data**, making it unstable and prone to errors.

3. **Q:** Name two primary methods used to mitigate overfitting in a single decision tree.

**A: Pruning** and **Ensemble Methods** (e.g., Bagging, Random Forests).

4. **Q:** What is the main strategy for improving the performance of weak learner models like decision trees?

**A: Combine multiple weak learners** to form a stronger, more stable predictor.

5. **Q:** What is the term for combining multiple trees to form a single prediction?

**A: Ensemble Learning.**

6. **Q:** In regression, how is the final prediction determined in an ensemble of trees?

**A:** By **averaging the predictions** from all individual trees.

7. **Q:** In classification, how is the final prediction determined in an ensemble?

**A:** By **majority vote** among the predictions of all trees.

8. **Q:** Name two ensemble-based strategies mentioned in the notes.

**A: Bagging** and **Boosting**.

9. **Q:** What does Bagging stand for?

**A: Bootstrap Aggregating.**

10. **Q:** What is the goal of the Bootstrap method in Bagging?

**A:** To generate multiple datasets by **sampling with replacement** to reduce variance and improve stability.

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## II. Bootstrap Aggregating (Bagging) (11–20)

11. **Q:** What sampling technique is used to generate datasets in Bagging?

**A: Bootstrap sampling** (sampling **with replacement**).

**12. Q:** What two key properties define bootstrap sampling?

**A:** **Random selection** and **replacement** (each datapoint can appear multiple times).

**13. Q:** In ML, what does "Aggregating" refer to?

**A:** **Combining predictions** from all bootstrapped models (average for regression, majority vote for classification).

**14. Q:** Formula for the final Bagged prediction ( $\hat{f}_{\text{bag}}(\vec{x})$ ) in regression?

**A:**

$$\hat{f}_{\text{bag}}(\vec{x}) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{(b)}(\vec{x})$$

where (B) is the total number of trees.

**15. Q:** How is the Bagged prediction determined for classification?

**A:** By **majority vote** among the predictions of the (B) trees.

**16. Q:** Fundamental difference between Bagging and Random Forest?

**A:** **Random Forest** adds **feature randomness** at each split, in addition to bootstrap sampling.

**17. Q:** Probability that a datapoint is not selected in any of the (n) draws as (n  $\rightarrow \infty$ )?

**A:** ( $P \rightarrow e^{-1} \approx 0.368$ ).

**18. Q:** Approximately what percentage of datapoints are never selected in a single bootstrap sample?

**A:** Around **36.8%**.

**19. Q:** What does (B) represent in Bagging?

**A:** The **number of trees/models** in the ensemble.

**20. Q:** Why is combining multiple predictions essential in Bagging?

**A:** It **reduces variance** and produces more stable and accurate predictions.

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### III. Out-of-Bag (OOB) Estimation (21–30)

**21. Q:** What does OOB stand for?

**A:** **Out-of-Bag**.

**22. Q:** What traditional technique is avoided using OOB estimation?

**A:** **Separate validation set** or cross-validation.

**23. Q:** In OOB, a data point is classified using only trees in which it was **not selected** during what process?

**A:** The **bootstrap sampling/training** process.

**24. Q:** For a datapoint (i), what are the trees that did not use it called?

**A:** The **OOB set**, denoted as  $\{\mathcal{B}_i\}$ .

**25. Q:** Formula for the average OOB prediction for  $\vec{x}_i$ ?

**A:**

$$\hat{f}_{\text{OOB}}(\vec{x}_i) = \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} \hat{f}^{(b)}(\vec{x}_i)$$

**26. Q:** OOB error estimate is averaged over what set of points?

**A:** Over **all datapoints** in the training dataset.

**27. Q:** Why are OOB data points considered “unseen” for their respective trees?

**A:** They were **not included in the bootstrap sample** used to train that tree.

**28. Q:** How is the optimal number of trees (B) determined using OOB?

**A:** By **tracking OOB error** as (B) increases and selecting (B) that **minimizes OOB error**.

**29. Q:** What does the optimization process seek to minimize for OOB performance?

**A:** The **average OOB prediction error** (MSE for regression, misclassification rate for classification).

**30. Q:** Why is OOB error computation computationally efficient?

**A:** Because it uses **already trained trees** without requiring a separate validation or cross-validation process.